

Deep Deep Learning

Graduate student-led research seminar | Semester: TBD | Course number: TBD

Instructor: Assaf Shocher. Course contact and office hours: TBD.

Course description

Major ideas in deep learning, studied through foundational material, a central paper and recent developments. The candidate pool includes NTK, Mamba, in-context learning, test-time training, hypernetworks, world models, online continual learning, model merging, interpretability, new generative frameworks, graph neural networks, KANs, and the theory of why training works.

Topic selection

The 23 topics in this syllabus are candidates, not weekly assignments. Students and the instructor choose which topics enter the final 13-week schedule. Unchosen topics are not assigned. Selection, order and presenter assignments: TBD. Each selected topic gets two academic hours.

Learning objectives

Explain a mathematical mechanism; identify its assumptions; assess an experiment that tests an explanation; connect recent research to older ideas; formulate a precise unresolved question.

Prerequisites & logistics

Working knowledge of deep learning, linear algebra, probability, calculus and optimization. Formal prerequisites, credits, enrollment, meeting time, room and delivery mode: TBD. Dates are not yet assigned. Flow matching is assumed background from the base course.

Seminar format & assessment

Students lead presentations and discussion. Each session centers on one main paper, with selected foundation and companion material. Presenters read the main paper and prepare one derivation, one decisive experiment and one unresolved question. No exam. Grading components and weights, attendance requirements and written submissions: TBD.

Shared pre-reading

All students complete one required pre-reading for each selected meeting. Presenters choose it in consultation with the instructor and announce it one week before class: a tutorial, a short paper or clearly specified sections taking 30-45 minutes. Include two guiding questions; each student arrives with one question or point of confusion. The main paper is optional for the audience unless chosen as the shared pre-reading. Specific pre-readings, slide deadlines and coordination arrangements: TBD.

Research practice & communication

Cite papers, figures, code and other sources; distinguish the authors' claims from your own analysis. Formal academic-integrity and AI-use policies, course forum and accessibility arrangements: TBD.

Weekly schedule

Week numbers only. Topic selections and student presenters are TBD. The candidate pool below is larger than the final course.

Week	Selected topic	Presenter
1	TBD	TBD
2	TBD	TBD
3	TBD	TBD
4	TBD	TBD
5	TBD	TBD
6	TBD	TBD
7	TBD	TBD
8	TBD	TBD
9	TBD	TBD
10	TBD	TBD
11	TBD	TBD
12	TBD	TBD
13	TBD	TBD

Suggested meeting structure

Two academic hours per selected topic. Assuming two 45-minute academic hours: 15 minutes foundations, 35 minutes mechanism and derivation, 15 minutes evidence, 25 minutes discussion. Adjust the split to the local academic-hour convention.

Choosing a manageable scope

Each candidate has one reading bundle and a central question. Build around its main paper, using selected foundation and companion sections to explain and assess the mechanism. Develop one derivation and one decisive experiment carefully. Candidate numbers are references for discussion, independent of assigned week numbers.

Candidate topic pool

Choose a numbered candidate for each selected session. Each reading bundle connects one main paper to its foundations and recent developments. Use selected sections to keep the central question within two academic hours. Topic numbers are independent of the weekly schedule. Shared pre-readings and selected sections for all candidates: TBD.

Kernels, feature learning & optimization

01 · NTK and feature learning

When can a neural network be understood as a kernel method, and when do its features change?

Background: Lee et al., [Wide Neural Networks of Any Depth Evolve as Linear Models Under Gradient Descent](#) (2019), selected sections on Taylor linearization and squared-loss dynamics. Review Jacobians, Gram matrices and gradient flow.

Main paper: Jacot, Gabriel & Hongler, [Neural Tangent Kernel: Convergence and Generalization in Neural Networks](#) (2018).

Recent development & discussion: Haas et al., [On the Surprising Effectiveness of Large Learning Rates under Standard Width Scaling](#) (2025).

Presentation focus: Derive function-space gradient flow and its solution for squared loss, then compare a finite network with its linearization. Use the recent paper to examine why width, parameterization, learning rate and the choice of loss matter for claims about feature learning.

02 · muP, scaling and hyperparameter transfer

How should initialization and updates scale as networks become wider or deeper?

Background: Yang & Hu, [Tensor Programs IV: Feature Learning in Infinite-Width Neural Networks](#) (2021), selected sections on parameterization and feature learning. Review the NTK linearized regime as contrasting background; a separate NTK seminar is optional.

Main paper: Yang et al., [Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer](#) (NeurIPS 2021; arXiv 2022).

Recent development & discussion: Dey et al., [Don't Be Lazy: CompleteP Enables Compute-Efficient Deep Transformers](#) (2025). Optional theory: Chen, Yang, Zhao & Gu, [Global Convergence and Rich Feature Learning in L-Layer Infinite-Width Neural Networks under \$\mu\$ Parametrization](#) (ICML 2025). Read the assumptions and result statement; this infinite-width analysis does not establish unrestricted finite-network SGD convergence.

Presentation focus: Work through activation and update scaling in a small multilayer network, and inspect a learning-rate transfer experiment. Discuss the additional constraints introduced by depth; width transfer alone does not establish depth transfer or effective nonlinear learning.

03 · Geometry of optimization: Muon

How does the norm used to measure an update change the direction of steepest descent?

Background: Bernstein & Newhouse, [Old Optimizer, New Norm: An Anthology](#) (2024), selected sections on steepest descent, dual norms and matrix operator norms.

Main paper: Liu et al., [Muon Is Scalable for LLM Training](#) (2025).

Recent development & discussion: Kovalev, [Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization](#) (2025).

Presentation focus: Derive the spectral-norm-constrained linearized update using the SVD and explain its polar factor and Newton-Schulz approximation. Examine a scaling experiment and distinguish the geometric argument from implementation choices, weight decay and empirical comparisons with AdamW.

04 · Why training works: convergence and loss landscapes

Which assumptions let us prove that training reaches a global minimum, and how close are those proofs to practical networks?

Scope for one session: Two academic hours: derive PL-based loss decay and examine one trajectory-wise convergence argument for a restricted neural-network model. Concentrate on the assumptions that make the proof work.

Background: Karimi, Nutini & Schmidt, [Linear Convergence of Gradient and Proximal-Gradient Methods Under the Polyak-Lojasiewicz Condition](#) (2016), the PL inequality and gradient-flow loss decay. Historical context: Choromanska et al., [The Loss Surfaces of Multilayer Networks](#) (2015; 2014 preprint), the spin-glass analogy and its assumptions. Optional contrast: Soudry & Hoffer, [Exponentially vanishing sub-optimal local minima in multilayer neural networks](#) (ICLR 2018), volumes of regions containing minima versus a training trajectory.

Main paper: Dana, Pillaud-Vivien & Bach, [Convergence of Shallow ReLU Networks on Weakly Interacting Data](#) (NeurIPS 2025). Gradient flow in finite shallow ReLU networks, weakly interacting inputs and a particular asymmetric randomized initialization.

Recent development & discussion: Petit, Poon & Peyré, [On the global convergence of gradient flow for wide shallow models beyond homogeneous nonlinearities](#) (2026). The result is conditional: if the studied Wasserstein gradient flow converges, its limit is globally optimal. Watch list: Soudry's [publication page](#) lists Global Convergence in Deep Networks via the Second Law of Thermodynamics (NeurIPS 2026); a linked manuscript was unavailable when this list was checked, so reading and claims remain TBD.

Presentation focus: Derive loss decay from a PL inequality, then explain how the main paper maintains a trajectory-wise condition. Inspect its data, initialization, width and gradient-flow assumptions. Separate landscape geometry, reaching a minimum, selecting a solution and generalization. These results make concrete progress; they do not supply a general guarantee for finite deep networks trained with SGD or Adam.

05 · Optimization at the edge of stability

Why can training improve near the usual gradient-descent stability threshold?

Scope for one session: Two academic hours: one quadratic stability calculation, self-stabilization and the averaged dynamics of gradient descent.

Background: Quadratic losses, Hessian eigenvalues and the gradient-descent stability threshold $2 / \text{learning rate}$. Cohen et al., [Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability](#) (ICLR 2021), short-term loss oscillations and longer-term progress.

Main paper: Cohen, Damian, Talwalkar, Kolter & Lee, [Understanding Optimization in Deep Learning with Central Flows](#) (ICLR 2025). Start with the gradient-descent case and the authors' [illustrated explanation](#).

Recent development & discussion: Berthier, [The Multiple Timescales of Gradient Descent on the Edge of Stability: A Perturbative Derivation of the Central Flow](#) (2026), selected introduction and toy example. This formal asymptotic derivation does not provide remainder bounds or a convergence proof.

Presentation focus: Derive the quadratic stability threshold, examine self-stabilization, and show how averaging oscillations produces a smoother trajectory description. Use one example to compare the discrete iterates and central flow. Explain what this reveals about optimization dynamics without treating it as a general global-convergence theorem.

Sequence models & reasoning

06 · State-space models and Mamba

How do recurrence, selectivity and structured matrices produce an efficient sequence model?

Background: Gu, Goel & Ré, [Efficiently Modeling Long Sequences with Structured State Spaces](#) (S4, 2021/22), selected sections on discretization and recurrence. Gu & Dao, [Mamba: Linear-Time Sequence Modeling with Selective State Spaces](#) (2023), selected sections on input-dependent dynamics.

Main paper: Dao & Gu, [Transformers Are SSMS: Generalized Models and Efficient Algorithms Through Structured State Space Duality](#) (Mamba-2, 2024).

Recent development & discussion: Lahoti et al., [Mamba-3: Improved Sequence Modeling Using State Space Principles](#) (2026).

Presentation focus: Unroll an SSM into a structured matrix and explain the restricted attention-SSM duality used by Mamba-2. Examine a state-tracking experiment from Mamba-3 and distinguish algorithmic complexity from measured hardware efficiency.

07 · In-context learning as an algorithm

How can a forward pass perform learning without changing the network's parameters?

Background: von Oswald et al., [Transformers Learn In-Context by Gradient Descent](#) (2023), selected sections on the linear-attention construction. Review least squares, ridge regression and Bayesian prediction for linear models.

Main paper: Akyürek et al., [What Learning Algorithm Is In-Context Learning? Investigations with Linear Models](#) (2023).

Recent development & discussion: Wurgajt et al., [In-Context Learning Strategies Emerge Rationally](#) (2025).

Presentation focus: Work through one regression-learning construction and compare gradient-descent, ridge and Bayesian predictors on a controlled task distribution. Separate existence constructions and observed behavior in trained synthetic-task models from claims about general language models.

08 · Test-time training (TTT)

What should a model learn at inference, and which objective makes those updates useful?

Scope for one session: Two academic hours: develop a recurrent TTT update, then examine end-to-end long-context training. Original distribution-shift TTT supplies the motivation and history; IT3 is supplementary reading.

Background: Sun, Wang, Liu, Miller, Efros & Hardt, [Test-Time Training with Self-Supervision for Generalization under Distribution Shifts](#) (ICML 2020), selected introduction and auxiliary-task formulation. Ba et al., [Using Fast Weights to Attend to the Recent Past](#) (2016), fast versus slow weights and associative memory.

Main paper: Sun et al., [Learning to \(Learn at Test Time\): RNNs with Expressive Hidden States](#) (2024). A model is the recurrent hidden state, and self-supervised gradient updates are the state transition.

Recent development & discussion: Tandon et al., [End-to-End Test-Time Training for Long Context](#) (2025). Next-token updates compress context into weights, with a meta-learned initialization. Supplementary adaptation example: Durasov, Shocher, Oner, Chechik, Efros & Fua, [IT3: Idempotent Test-Time Training](#) (ICML 2025); [implementation](#).

Presentation focus: Start with the original 2020 adaptation problem, then work through a TTT-Linear update and its recurrent computation. Examine how end-to-end next-token training compresses context into weights, and explain the training and inference costs. IT3 provides a supplementary consistency-based adaptation example with its own training setup.

09 · RL for reasoning with verifiable rewards

Does reward-based training discover successful reasoning behavior or change which existing behavior is sampled?

Background: [Spinning Up: Part 3-Introduction to Policy Optimization](#), selected sections on the log-derivative policy gradient and baselines. Reserve 25-30 minutes to introduce policies, sampled completions, rewards and group-relative advantages; prior RL coursework is not assumed.

Main paper: DeepSeek-AI et al., [DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning](#) (2025), focusing on R1-Zero and the learning objective.

Recent development & discussion: Yue et al., [Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?](#) (2025). Counterpoint: Liu et al., [ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models](#) (2025).

Presentation focus: Derive the policy-gradient identity before introducing group-relative advantages, clipping and KL regularization. Compare pass@k evidence under the papers' different training and evaluation settings; finite sampling cannot establish that a base model assigns zero probability to a solution.

10 · Recurrent depth and latent reasoning

Can a model gain reasoning ability by repeatedly applying the same layers?

Background: Dehghani et al., [Universal Transformers](#) (ICLR 2019; 2018 preprint). Shared layers, recurrence over depth and adaptive computation.

Main paper: Geiping et al., [Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach](#) (NeurIPS 2025).

Recent development & discussion: Jolicoeur-Martineau, [Less is More: Recursive Reasoning with Tiny Networks](#) (2025). A short comparison with recursive refinement on puzzles.

Presentation focus: Trace one recurrent block, explain how it is trained across iteration counts, and assess performance against inference compute. Use tiny recursive models as a comparison. Distinguish iteration count, parameter count and demonstrated reasoning performance.

Representations, world models & continual learning

11 · JEPA and predictive world models

When does prediction in representation space support understanding, dynamics and planning?

Background: Assran et al., [Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture](#) (I-JEPA, 2023), selected sections on masking, target representations and collapse prevention.

Main paper: Assran et al., [V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning](#) (2025).

Recent development & discussion: Maes et al., [LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels](#) (2026).

Presentation focus: Trace the representation-prediction objective and action-conditioned planning loop, then analyze a planning result and its data requirements. Contrast target-network stabilization with regularizing the latent distribution; representation quality and accurate action-conditioned dynamics are separate questions.

12 · Online continual learning: stability, plasticity and replay

How can a network keep adapting to a changing data stream while retaining useful knowledge?

Scope for one session: Two academic hours: derive OCAR's replay-constrained update, inspect one streaming experiment, and use CLA to connect prediction preservation with self-supervised representation alignment. A one-pass incoming stream can allow replay and bounded repeated updates on the current minibatch; it does not imply exactly one gradient step per observation.

Background: Chaudhry et al., [On Tiny Episodic Memories in Continual Learning](#) (2019): experience replay with limited memory. Kirkpatrick et al., [Overcoming Catastrophic Forgetting in Neural Networks](#) (PNAS 2017): elastic weight consolidation and Fisher-weighted protection. Optional mathematical background: Martens & Grosse, [Optimizing Neural Networks with Kronecker-factored Approximate Curvature](#) (ICML 2015), selected sections introducing the Fisher and one K-FAC block.

Main paper: Urettini & Carta, [Online Curvature-Aware Replay: Leveraging 2nd Order Information for Online Continual Learning](#) (ICML 2025). Focus on Sections 3-5: formulate the streaming update, constrain changes to predictions on replay data, and derive the curvature-aware step.

Recent development & discussion: Cignoni et al., [CLA: Latent Alignment for Online Continual Self-Supervised Learning](#) (CoLLAS 2026): compare representation alignment with labeled prediction constraints. Conceptual companion: Dohare et al., [Loss of Plasticity in Deep Continual Learning](#) (Nature 2024), which distinguishes forgetting old knowledge from losing the ability to learn new tasks; its broader continual-learning experiments are not all single-pass online protocols.

Presentation focus: Derive the local KL/Fisher constraint and the preconditioned update on a small example. Explain damping, memory representativeness and the difference between forgetting and loss of plasticity. Evaluate accuracy during the stream as well as at its end. Check what information each experiment provides: OCAR hides task boundaries, but its class-incremental replay-Fisher weighting uses the number of new classes per experience.

13 · Platonic representations and unpaired alignment

Does shared representation geometry let us align independently trained models without paired alignment examples?

Scope for one session: Two academic hours: representation similarity, the Platonic hypothesis and one concrete unpaired alignment algorithm. Develop the Procrustes step on a small example; explain the full matching procedure without deriving every optimization subproblem.

Background: Kornblith et al., [Similarity of Neural Network Representations Revisited](#) (ICML 2019), centered kernel alignment. Huh et al., [Position: The Platonic Representation Hypothesis](#) (ICML 2024), the hypothesis and supporting experiments. Review an orthogonal Procrustes alignment using the SVD.

Main paper: Schnaus, Dagès, Cremers, Wang & Isola, [Shared Geometry As A Rosetta Stone: Cross-Modal Alignment Without Paired Data](#) (October 2026 preprint); the authors' [interactive project](#) illustrates coarse cross-modal alignment without pairs at the alignment stage.

Recent development & discussion: Gröger, Wen & Brbic, [Revisiting the Platonic Representation Hypothesis: An Aristotelian View](#) (ICML 2026). Use the critique to examine which definitions and comparisons support claims about shared representations.

Presentation focus: Treat Rosetta as a practical test of a consequence of the Platonic hypothesis. Explain its coarse geometric initialization and alternating assignment / Procrustes updates, then examine the broad semantic structure it recovers and the finer details it misses. Paired samples are still used for evaluation and the CKA analysis of when alignment succeeds. Distinguish useful geometric correspondence from universal convergence or canonical features.

Weight space & interpretability

14 · Hypernetworks and learning in weight space

How can a neural network generate, read or edit another network's weights while respecting their symmetries?

Scope for one session: One seminar of two academic hours. The core is permutation symmetry and one weight-space expressivity result. HyperNetworks supplies the generation idea; SHINE supplies one current application. Doc-to-LoRA is supporting reading.

Background: Ha, Dai & Le, [HyperNetworks](#) (ICLR 2017), a network generating another network's parameters. Navon, Shamsian, Achituve, Fetaya, Chechik & Maron, [Equivariant Architectures for Learning in Deep Weight Spaces](#) (ICML 2023), selected permutation action and DWS-layer construction; the authors' [tutorial](#) is an accessible introduction. Hu et al., [LoRA: Low-Rank Adaptation of Large Language Models](#) (2021/22), selected low-rank update formulation.

Main paper: Dayan, Eitan & Maron, [On the Expressive Power of Permutation-Equivariant Weight-Space Networks](#) (ICML 2026). Focus on the approximation framework and one universality result for fixed-architecture MLP weight spaces: distinguish properties of parameterizations from properties of the functions those parameters realize.

Recent development & discussion: Liu, Wang, Mao, Gelberg, Maron & Zhang, [SHINE: A Scalable In-Context Hypernetwork for Mapping Context to LoRA in a Single Pass](#) (ICML 2026), a contemporary application that converts context into adapters. Supporting application reading: Charakorn et al., [Doc-to-LoRA: Learning to Instantly Internalize Contexts](#) (ICML 2026), amortized context distillation.

Presentation focus: Derive the hidden-neuron permutation action for a small MLP and construct a simple equivariant weight-processing layer. Explain why arbitrary flattening obscures this structure. Work through one proof sketch connecting canonicalization to universality under a general-position assumption, then inspect a degenerate counterexample. End with a short SHINE example contrasting direct adapter generation with iterative parameter updates. SHINE does not instantiate the DWS construction; keep the MLP expressivity results separate from the empirical claims about LLM adaptation.

15 · Model merging and pretrained weight neighborhoods

Which useful capabilities lie near a pretrained checkpoint, and how can we recover or combine them?

Scope for one session: Two academic hours: introduce task vectors and weight averaging, then center the seminar on Neural Thickets and the difference between combining weights and ensembling predictions.

Background: Ilharco et al., [Editing Models with Task Arithmetic](#) (2023), task vectors relative to a shared pretrained checkpoint. Wortsman et al., [Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time](#) (ICML 2022), weight averaging versus prediction ensembles. Selected theory: Ortiz-Jimenez, Favero & Frossard, [Task Arithmetic in the Tangent Space: Improved Editing of Pre-Trained Models](#) (2023), the linearized model and weight disentanglement.

Main paper: Gan & Isola, [Neural Thickets: Diverse Task Experts Are Dense Around Pretrained Weights](#) (ICML 2026). RandOpt samples Gaussian weight perturbations, evaluates candidates on task data, selects experts and ensembles their predictions. See the authors' [illustrated project](#).

Recent development & discussion: Cendra et al., [Can Bayesian Optimization Efficiently Find a Strong Single Expert in Neural Thickets?](#) (2026), a follow-up in low-dimensional search settings. Empirical comparison: Hitit, Gırrbach & Akata, [A Systematic Study of In-the-Wild Model Merging for Large Language Models](#) (TMLR 2026; 2025 preprint), heterogeneous checkpoints and conflicting objectives.

Presentation focus: Use function-space linearization to explain the intuition behind task-vector addition. Then develop expert density and diversity around a pretrained checkpoint, including the probability of finding a useful perturbation after repeated sampling. Compare weight averaging, expert selection and prediction ensembling, accounting for task-data evaluations and inference cost. These neighborhood results do not establish that random initialization or unrestricted gradient descent succeeds.

16 · Mechanistic interpretability: from features to circuits

How do interpretable features support a circuit explanation, and how can we test that explanation causally?

Scope for one session: Two academic hours: introduce superposition and sparse features, then develop one circuit-tracing case study and causal intervention.

Background: Elhage et al., [Toy Models of Superposition](#) (2022), one small example of sparse features sharing dimensions.

Selected background: Elhage et al., [A Mathematical Framework for Transformer Circuits](#) (2021), the residual stream; Templeton et al., [Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet](#) (2024), the sparse-autoencoder objective and one feature illustration.

Main paper: Ameisen et al., [Circuit Tracing: Revealing Computational Graphs in Language Models](#) (2025).

Recent development & discussion: Kamath et al., [Tracing Attention Computation Through Feature Interactions](#) (2025). Limitation reading: Leask et al., [Sparse Autoencoders Do Not Find Canonical Units of Analysis](#) (2025), a caution about treating extracted features as canonical units.

Presentation focus: Use a toy example to explain superposition and introduce sparse feature extraction briefly. Center the presentation on how a replacement model produces an attribution graph, one circuit case study and a causal intervention. Distinguish sparse autoencoders from the tracing method's replacement model. Assess omitted computations and approximation error; interpreting a feature and validating a circuit are different steps.

Generative models

17 · One-step generative models: Drifting and IMLE

How can matching real and generated samples train a generator that produces a sample in one pass?

Scope for one session: One comparative seminar of two academic hours: derive Drifting as the central sample-matching mechanism, then compare IMLE's nearest-neighbor assignments and ROMS-IMLE's practical changes.

Background: Yujia Li, Swersky & Zemel, [Generative Moment Matching Networks](#) (ICML 2015), selected kernel/MMD material. Ke Li & Jitendra Malik, [Implicit Maximum Likelihood Estimation](#) (2018), the data-to-generated nearest-neighbor assignment and regression algorithm.

Main paper: Deng, He Li, Tianhong Li, Du & He, [Generative Modeling via Drifting](#) (2026). Estimate an attraction-repulsion field from real and generated samples and fit frozen drifted targets.

Recent development & discussion: Chirag Vashist & Ke Li, [ROMS-IMLE: A Minimalist Approach to Competitive Single-Step Generative Modelling](#) (2026). Compare nearest-neighbor assignment, robust losses and intermediate-resolution supervision with the drifting update. Optional theory: Esteban-Casadevall et al., [Kernel-Gradient Drifting Models](#) (2026), including Max Welling. The authors' [ROMS-IMLE method page](#) provides additional explanation and experiments.

Presentation focus: Introduce sample-based distribution matching, derive the drifting field and frozen-target regression, and study one toy experiment. Compare IMLE's data-to-generated matching and ROMS-IMLE's practical changes, including the direction of nearest-neighbor assignment, robust losses and intermediate-resolution supervision. Keep zero-drift and maximum-likelihood claims tied to their assumptions.

18 · Diffusion language models

How can denoising generate discrete text, and what changes when generation need not proceed left to right?

Background: Austin et al., [Structured Denoising Diffusion Models in Discrete State-Spaces](#) (2021), selected sections on transition matrices and absorbing-mask corruption.

Main paper: Sahoo et al., [Simple and Effective Masked Diffusion Language Models](#) (2024).

Recent development & discussion: Nie et al., [Large Language Diffusion Models](#) (LLaDA, 2025).

Presentation focus: Derive the masked-denoising objective and trace one sampling schedule, then compare refinement, parallelism and likelihood evaluation with autoregressive generation. The topic requires discrete corruption and language-specific evaluation beyond a continuous diffusion refresher.

Graphs, symmetry & neural architectures

19 · Symmetry, equivariance and canonicalization

Can we enforce symmetry by putting each input into a canonical orientation?

Scope for one session: Advanced mathematical candidate. A concrete rotation or permutation example is enough to motivate the main result within two academic hours.

Background: Cohen & Welling, [Group Equivariant Convolutional Networks](#) (ICML 2016). Group actions, equivariance and weight sharing.

Main paper: Dym, Lawrence & Siegel, [Equivariant Frames and the Impossibility of Continuous Canonicalization](#) (ICML 2024).

Recent development & discussion: Shumaylov et al., [Lie Algebra Canonicalization: Equivariant Neural Operators under arbitrary Lie Groups](#) (ICLR 2025; 2024 preprint). Selected introduction and one application.

Presentation focus: Start with a group-action example, show why a canonical choice can become discontinuous under the stated assumptions, and explain how weighted frames address continuity. Leave general topology proofs and non-compact Lie-group machinery for optional reading.

20 · Oscillatory neurons and synchronization

Can neural representations bind information through synchronization?

Scope for one session: Exploratory candidate connecting neural computation with dynamical systems, including work by Max Welling.

Background: Strogatz, [From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators](#) (2000). Sections 2 and 3.1-3.3 on coupled phases and synchronization (author-hosted PDF).

Main paper: Miyato, Löwe, Geiger & Welling, [Artificial Kuramoto Oscillatory Neurons](#) (ICLR 2025; 2024 preprint).

Recent development & discussion: Iqbal et al., [Spontaneous symmetry breaking and Goldstone modes for deep information propagation](#) (2026 preprint). A related perspective on symmetry and persistent information propagation, rather than a direct analysis of AKOrN.

Presentation focus: Derive the two-oscillator intuition, trace the generalized neural update, and study one object-discovery or reasoning experiment. Use the Goldstone paper for a brief connection to depth and memory; its physics derivations are optional.

21 · KANs: learnable functions and scientific discovery

When does learning the functions on network edges help more than learning ordinary weights?

Scope for one session: Two academic hours: the original architecture, one scientific example and one controlled comparison. Approximation-rate proofs are optional.

Background: Kolmogorov, [On the Representation of Continuous Functions of Several Variables as Superpositions of Continuous Functions of One Variable and Addition](#) (1957), theorem statement only. Review one-dimensional spline bases using the main paper's architecture section. Separate representation existence from efficient approximation and learnability.

Main paper: Liu et al., [KAN: Kolmogorov-Arnold Networks](#) (ICLR 2025; 2024 preprint), selected sections on learned edge functions, grid refinement and one function-fitting experiment.

Recent development & discussion: Liu et al., [KAN 2.0: Kolmogorov-Arnold Networks Meet Science](#) (2024), one scientific example. Essential comparison: Yu, Yu & Wang, [KAN or MLP: A Fairer Comparison](#) (2024). Recent optional theory: Kratsios, Kim & Furuya, [Approximation Rates in Besov Norms and Sample-Complexity of Kolmogorov-Arnold Networks with Residual Connections](#) (2025), assumptions and result statements.

Presentation focus: Build a small spline-based KAN and explain its compositional bias and possible formula recovery. Compare against an MLP with matched parameters or FLOPs and a spline-equipped MLP. Ask which benefit comes from the function basis and which comes from the architecture; the representation theorem alone does not establish a practical advantage.

22 · GNNs: message passing and graph transformers

What graph structure can a neural network distinguish, and what does global attention add?

Scope for one session: Two academic hours: message-passing expressivity, a concrete 1-WL limitation, and structural encodings in graph transformers.

Background: Sanchez-Lengeling et al., [A Gentle Introduction to Graph Neural Networks](#) (Distill 2021), a visual primer. Xu, Hu, Leskovec & Jegelka, [How Powerful are Graph Neural Networks?](#) (ICLR 2019), multiset aggregation, GIN and the 1-Weisfeiler-Leman expressivity ceiling.

Main paper: Stoll, Müller & Morris, [Generalizable Insights for Graph Transformers in Theory and Practice](#) (NeurIPS 2025), selected theory sections and one structural-encoding ablation.

Recent development & discussion: Ma et al., [Plain Transformers Can be Powerful Graph Learners](#) (TMLR 2026; 2025 preprint). Examine the graph-specific adaptations behind the title. Optional benchmark discussion: Bechler-Speicher et al., [Position: Graph Learning Will Lose Relevance Due To Poor Benchmarks](#) (ICML 2025).

Presentation focus: Work through a pair of graphs that 1-WL-style message passing cannot distinguish. Explain how structural or positional information changes the comparison with a graph transformer. Distinguish theoretical expressivity, measured accuracy and benchmark quality; leave the full higher-order hierarchy for optional reading.

23 · GNNs: over-squashing and graph geometry

How does graph connectivity limit the information a message-passing network can transmit?

Scope for one session: Two academic hours: a bottleneck example, over-squashing and one graph-propagation construction. Introduce expander connectivity intuitively; full group-theory and spectral proofs are further reading.

Background: Kipf & Welling, [Semi-Supervised Classification with Graph Convolutional Networks](#) (ICLR 2017), neighborhood propagation. Alon & Yahav, [On the Bottleneck of Graph Neural Networks and its Practical Implications](#) (ICLR 2021), information bottlenecks and over-squashing.

Main paper: Wilson, Bechler-Speicher & Velickovic, [Cayley Graph Propagation](#) (2025), an expander-based propagation template.

Recent development & discussion: Liang et al., [Mitigating Over-Squashing in Graph Neural Networks by Spectrum-Preserving Sparsification](#) (ICML 2025), a comparison of graph modification and propagation costs.

Presentation focus: Use a bottleneck graph to distinguish over-squashing from over-smoothing and from limited graph-distinguishing expressivity. Explain expander connectivity, inspect one propagation experiment, and account for the cost of modifying graph connectivity.

Course website: [Deep Deep Learning](#)